A systematic evaluation of Anti-CD19 CAR T-Cell therapy for the potential treatment of systemic lupus erythematosus: A scoping review

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Abstract

Systemic Lupus Erythematosus (SLE) is a chronic, autoimmune disease characterized by the production of autoantibodies leading to tissue and organ damage. Current treatments fail to selectively target autoreactive immune cell populations and also lead to broad immunosuppression. Given the central role of B cells in SLE, there has been an increasing interest in using CAR-T therapy for the treatment of SLE by targeting autoreactive B cell populations. A PRISMA scoping review was conducted to assess the efficacy and remission rates for CAR T-cell therapy, an emerging immunotherapy which can be genetically engineered to detect CD19 co-receptors on B cells and eliminate them. Eleven studies consisting of 73 patients from both case reports and pilot studies sourced from PubMed and the ACR were analyzed. Our study showcases that CD19 CAR T-cell therapy achieves remarkable outcomes by depleting circulation of B cell populations, ablating autoreactive B cell populations, and inducing remission. Data revealed that most patients achieve remission by 3 months with normalization of C3 and C4 levels and no serious adverse complications in any of the patients that completed the minimum 3 months follow-up. This study shows that CAR T-cell therapy, autologous or allogeneic, is a potentially effective and feasible treatment option for patients with severe SLE with progressing organ damage that are refractory to the currently available treatment options. However, adequate clinical studies in this field remain limited, underscoring the need for large-scale, randomized clinical trials to better understand the safety and efficacy of CD19 CAR T-cell therapy.

Keywords: systemic lupus erythematosus (SLE), autoimmunity, CD19, chimeric antigen receptor t cell (CAR-T), remission

Introduction

Systemic Lupus Erythematosus (SLE) is a chronic, systemic autoimmune disease associated with the production of autoantibodies and immune complex-mediated inflammatory damage to various tissues and organs in the body. This condition manifests in a wide variety of clinical symptoms such as fatigue, joint pain, small blood vessel damage, seizures, dermatological facial rashes, kidney disease, cardiovascular problems such as Libman-Sacks endocarditis, and neuropsychiatric disorders [1, 2]. While SLE can affect anyone, it predominantly occurs in women during the childbearing years (ages 15-44) and in Hispanic, Asian, and African-American populations [1, 2]. Despite a variety of treatment options such as antimalarial drugs with anti-inflammatory properties, immunosuppressants, and monoclonal antibodies, these treatments still fail to selectively target autoreactive immune cells resulting in poor long-term prognosis for SLE patients [3, 4]. In recent years, Chimeric Antigen Receptor (CAR) T-Cell therapy has emerged as a potential new alternative treatment for refractory SLE patients.

Pathogenesis and Immune Mechanisms of SLE

The pathophysiology of SLE is complex and is characterized by the breakdown of immunological self-tolerance in both the innate and adaptive systems (Figure 1). Our current understanding is that environmental triggers such as viral/bacterial infections, ultraviolet light, and drugs can trigger defective innate and adaptive immune responses in genetically susceptible individuals [5, 6]. Initially, the impaired or delayed clearance of intracellular molecules released from apoptosis or NETosis in tissues act as potent autoantigenic material [1, 7].

Autoimmunity in the innate system can be achieved through Toll-like receptor (TLR) dependent activation which involves endosomal and defective cell-membrane bound TLRs such as TLR 7 and 9 binding to self-antigens in macrophages and plasmacytoid Dendritic cells (pDC) leading to the activation of molecular pathways that release inflammatory cytokines such as interferon-beta (IFN-β) and interferon-alpha (IFN-a) which are class 1 interferons [8, 9, 10]. The production of cytokines stimulates lymphocytes, dendritic cells (DCs) and natural killer (NK) cells leading to the breakdown of self-tolerance (Figure 1). Alternatively, TLR independent activation occurs when auto-antigens trigger RNA and DNA sensors activating IRF3 and NF-κB leading to neutrophil activation, triggering inflammation, and activating the complement system [8, 11, 12].

In the adaptive immune system, autoimmunity is enabled by the breakdown of central and peripheral tolerance mechanisms, the natural screening processes for T and B cells. Initially, clonal depletion and receptor revision mechanisms during central tolerance, which occurs in the early stages of B cell development, become dysregulated which allows high-affinity autoreactive clones to enter the periphery. In the peripheral tissues, which includes the secondary lymphoid organs, mechanisms such as suppression by Treg cells and apoptosis are also defective in SLE. This systematic failure of both tolerance mechanisms leads autoreactive immune cells to expand rather than be eliminated [13]. Additionally, antigen-presenting cells (APCs) present apoptotic debris and cellular self-antigens to these autoreactive T and B-lymphocytes leading to the production of cytokines and altered regulatory T-cell production. Further damage to multiple organ systems leads to more autoantigen circulation, and epitope spreading of new autoreactive B cells becoming activated (Figure 1). The interaction of T-cells and B cells via CD40L ligand and cytokines along with dysregulated signals received from both TLRs and B-cell receptors

activates autoreactive B cells by promoting their proliferation and differentiation into high-affinity autoantibody secreting plasma blasts and short-lived plasma cells (SLPC), both which are both acute antibody producing short-lived B cells [8, 12]. Autoantibodies such as anti-DNA, anti-Sm, anti-Ro, and Anti-Lo are pathogenic and cause organ damage by immune complex deposition and neutrophil activation [1, 8]. Simultaneously, immunoregulatory mechanisms such as the production of Treg cells and autoreactive B cell neutralization mechanisms remain impaired [8].

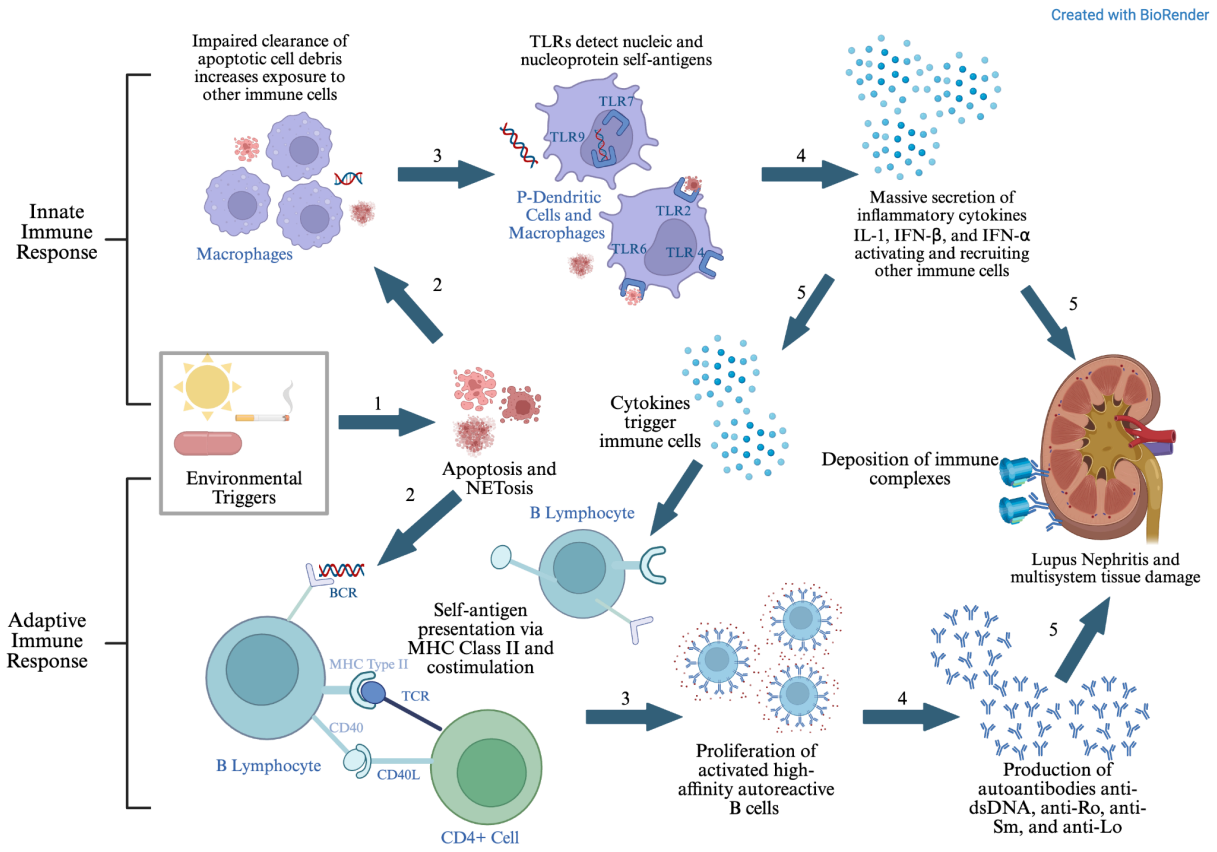

Figure 1: Overview of SLE mechanism. Initially, Environmental triggers lead to a cascade of immune responses that release cytokines and activate B and T cells leading to autoantibodies that damage tissue. On the innate side, macrophages and pDCs detect self-antigens and secrete

inflammatory cytokines leading to autoimmune activation. On the adaptive side, CD4+ cells present self-antigens which triggers B cells to proliferate into high-affinity autoantibody-secreting B cells. Together, this results in deposition of immune complexes leading to tissue and organ damage.

Current Treatment Landscape

Assessment Scores and Remission Status

The goal of SLE treatment is to achieve sustained remission. The DORIS (Definitions of Remission in SLE) criteria is a globally accepted and standardized definition of remission in SLE consisting of: SLEDAI (SLE Disease Activity Index) score of 0, Evaluator's Global Assessment score of <0.5 (on scale of 0-3), can be on an antimalarial drug, prednisolone 5 mg/day (or equivalent glucocorticoid steroid) or less, and/or on stable immunosuppressives including biologics [14]. The SLEDAI score is a clinical index for the assessment of Lupus disease activity consisting of 24 weighted clinical and laboratory variables of nine organ systems with neurologic, vascular, and musculoskeletal symptoms holding the highest weight. A score of 0 is associated with remission and indicates no disease activity while a score of >10 indicates severe disease activity (105 being the maximum theoretical SLEDAI score). A 4 point increase from baseline would be considered a flare up or sudden worsening of the disease [15].

Current Treatments and Limitations

Hydroxychloroquine, an antimalarial drug with anti-inflammatory properties, is the 1st line treatment for SLE. However, it is known to cause retinal toxicity which can lead to blindness absent close monitoring [16]. In addition, various immunosuppressive medications and

monoclonal antibodies such as rituximab, anifrolumab, and belimumab are also available as treatment options which are selected based on individual patient factors and affected organ systems [16]. While current treatments can help prevent severe organ damage and limit symptoms to improve patient care, many patients with SLE respond to medications poorly due to their broad immunosuppressive side effects, with many relapsing. One study pooled data from 5 randomized clinical trials and assessed patients on belimumab, one of few monoclonal antibody treatments approved for SLE, versus placebo once a month for a year. At week 52, 8% of participants on belimumab vs 6% of participants on placebo achieved remission [17]. Another study found that among patients currently living with SLE 26.3%, 51.0%, and 22.7% have mild, moderate, and severe disease, respectively with an average of 2.2, 3.7, and 4.2 flare-ups per year, respectively [18]. This data suggests that the overall efficacy with the currently available treatments in regards to being able to bring the disease under remission remains low.

CAR T-Cell Mechanisms

Overview

Given the central role of B cells in Lupus pathology, there has been a growing interest in therapies that target CD19, which are co-receptors on B cells, for the treatment for SLE [19]. One such therapy is CAR T-cell therapy, originally utilized in treatments of refractory cancers. CAR T-cell therapy involves removing a patient's own T cells, then genetically modifying them to express receptors that recognize specific antigens, harvesting the genetically modified T cells, and then reinfusing them back into the patient. This method is called “autologous CAR T-cell therapy (Figure 2A). Currently, all FDA approved CAR-T therapies utilize autologous T cells. However, due to the treatment’s highly individualized nature, it is time and resource intensive

with higher risks of failure during the manufacture and harvest of CAR T cells. Allogeneic CAR T therapy is an alternative approach involving the infusion of CAR T cells from healthy donors (Figure 2B). However, this method risks rejection from the recipient's immune system which results in graft-versus-host disease (GVHD) [38]. Overall, while this novel method has shown to be very effective in targeting and depleting specific blood cells for specific hematologic cancers, CAR T-cell therapy directed towards CD19 co-receptors in B cells has the potential to treat certain autoimmune diseases, such as in SLE [20 & 21].

Innovation and Developments

Currently undergoing several developments, five generations of CAR T-cell therapy currently exist (Figure 2C-D). The first generation of CAR T-cell contains a single-chain s (scFvs) which provides an extracellular antigen-binding region, which are small-sized proteins composed of the immunoglobulin heavy and light chain variable regions connected by a peptide linker and is the smallest unit of an antibody that can bind to antigens, fused to CD3zeta, an intracellular signaling domain [22]. T cell activation typically requires two signals - the first being triggered by the T-cell receptor (TCR) engaging with a peptide antigen anchored on the peptide-major histocompatibility complex (pMHC), and the second signal provided by costimulatory receptors such as CD28 [23]. The second generation of CAR T-cell builds on that logic by adding a CD28 co-receptor to the scFvs and CD3zeta complex providing both activation and co-stimulation. This modification to the original version allows for greater signaling, antigen-dependent activation, proliferation, and cytotoxic T cell effects [24]. Due to enhanced functionality, the superior second-generation CAR T have now largely replaced the outdated

first-generation CAR T-cells with the third through fifth generations being modified versions of the second generation adding additional co-stimulators [24].

Side Effects

However, CAR T-cell therapy has been associated with certain unique and serious side effects. One common side effect is cytokine release syndrome (CRS) which can be classified as a grade from 1 to 4 depending on severity. The majority of patients experience grade 1 CRS which is generally mild and consists mainly of fever. Less common are grades 3 to 4 CRS which can be life-threatening and requires full ICU care, tocilizumab treatment, high dose steroids, treatment for low blood pressure, and oxygen supplementation with the possibility of mechanical ventilation. Other possible side effects observed in cancer patients with CAR T-cell treatment are immune effector cells–associated neurotoxicity syndrome (ICANS) and prolonged cytopenia which can increase the risk of anemia and infections.

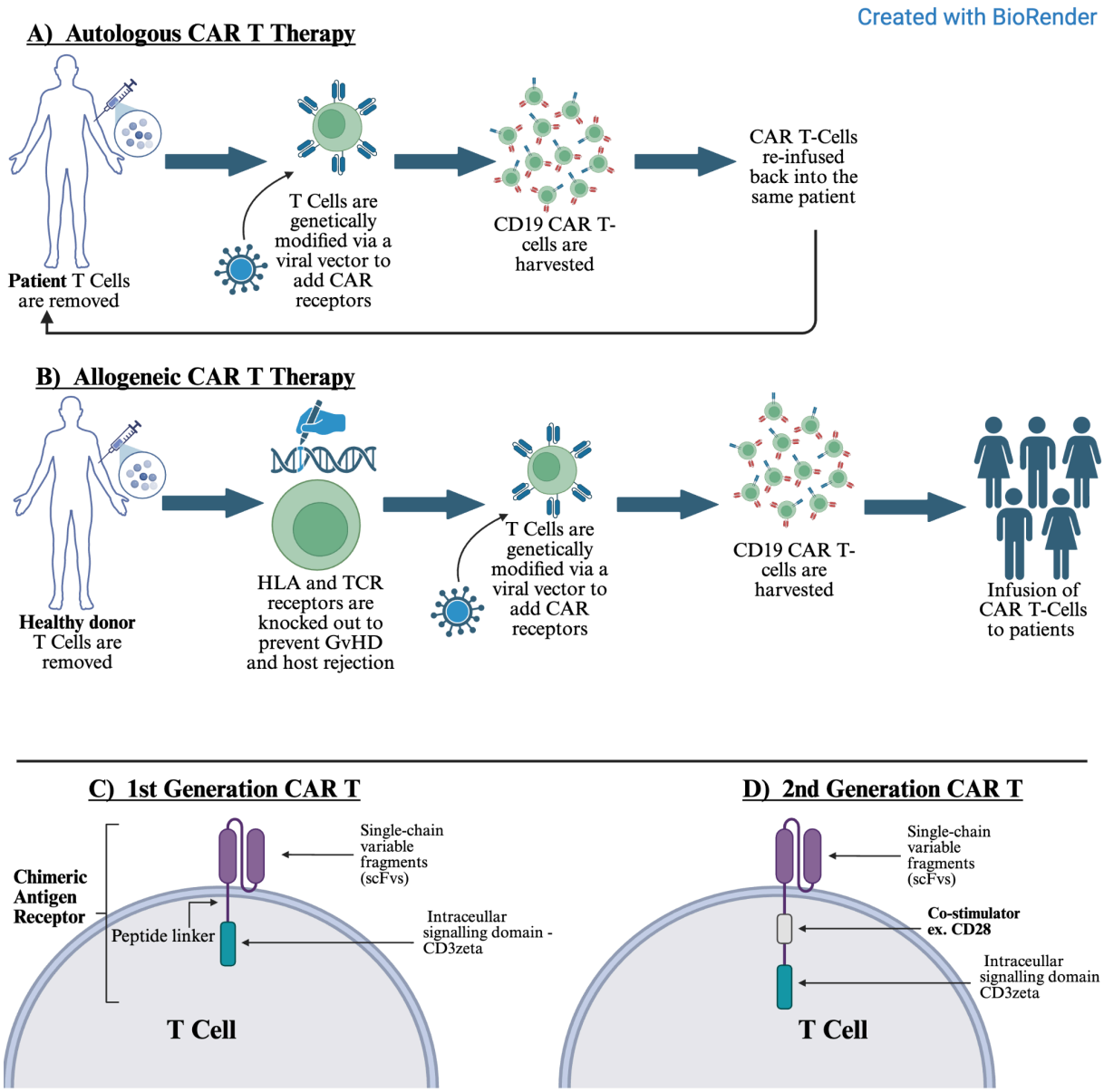

Figure 2: CAR T-Cell Therapy and Mechanisms. A) Autologous CAR T therapy involves the infusion of a patient's genetically modified T cells with CAR receptors back into the original patient. B) Allogeneic CAR T therapy involves the transfer of healthy donor's CAR T-cells into other patients C) The first generation of CAR T involves a simple CAR receptor consisting of a scFvs and CD3zeta D) The second generation of CAR T involves an added co-stimulator to allow for greater signaling and proliferation.

Method

This paper was conducted following the PRISMA 2020 guideline for reporting systematic reviews to ensure transparency and reproducibility of selection and inclusion of relevant studies [25]. Utilizing PubMed as well as the American College of Rheumatology (ACR) clinical database, 5 searches queries were run: “CAR T AND Lupus”, “CAR T AND SLE”, “Allogeneic AND Lupus”, “Allogeneic AND CAR T AND Lupus”, and “CAR T CD19 Lupus SLE”. For the PubMed database, “Clinical Study” and “Case Reports” filters were applied and publishing dates were limited to 5 years to ensure recency (Figure 3). Non-English articles and duplicate studies were removed. 31 total records were identified from the PubMed database and 63 records were identified from the ACR database. Inclusion criteria were (1) case studies involving patients with refractory SLE and (2) a follow-up at least 3 months post CAR-T with results. Excluded were case reports with no remission results available and data from patients who discontinued CAR-T therapy (3) studies published within 5 years before June 2025. 94 studies were screened and abstracts passed for the literature review if they were original research/clinical data from case studies done in human patients. 63 reports were excluded in the initial step of the screening process if they were solely focused on non-Lupus diseases or non-human animals. Finally, 5 studies were excluded because they were secondary literature such as a review or editorial. 15 were excluded for lack of significant relevance to the research question. Overall, 11 studies were included in the literature review as they were original case studies with unique data.

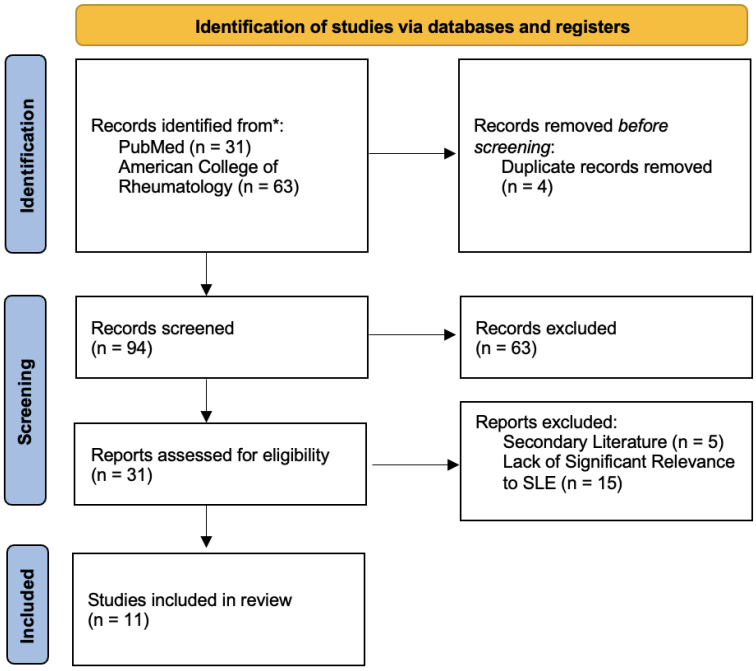

Figure 3. ***Flowchart of PRISMA Screening Procedure***

Results

SLEDAI data was derived from five studies with available information and was charted. Patients with higher baseline SLEDAI scores, which indicates more severe SLE, took longer on average to reach a score of zero after receiving CD19 CAR-T therapy with every single patient reaching remission (Figure 4A). With the exception of one patient who discontinued CD19 CAR T therapy, 71 out of 73 total patients reached remission, including those without SLEDAI data in Figure 4A [26-36]. Additionally, B cell counts fall drastically for most patients by eight days, an expected finding as B-cell aplasia develops and is maintained over a three-to-six-month period, followed by a gradual return to normal levels (Figure 4B). Further immunophenotyping of the B cells that returned were naive and showed non-class-switched B cell receptors. This indicates that the activated autoreactive CD19+ B cells, namely the plasma blasts and short-lived plasma cells, were eliminated and that the B cells that returned were “reset” back to the naive phase [29].

CAR T cell counts in the blood increase and peaks at about 10-20 days before gradually decreasing with levels remaining very low or non-detectable 3 months after infusion of CAR T, although the exact counts varied significantly with each patient (Figure 4C).

Importantly, during B-cell aplasia, serologic markers of SLE also fell. Auto-antibody anti-dsDNA counts are eliminated by 3 months and the majority of the patient's complement C3 and C4 levels rose to normal range levels indicating the deactivation of autoimmune attack complexes on organs (Figure 4D-F). For patients with proteinuria, the number of proteins in the urine which is one indicator of kidney disease also gradually fell over a 3-6 month period [26]. Serologic remissions were also paralleled by clinical remissions. The majority of signs and symptoms of SLE in patients such as kidney disease, skin rash, arthritis pain and fevers also improved in conjunction [26-36].

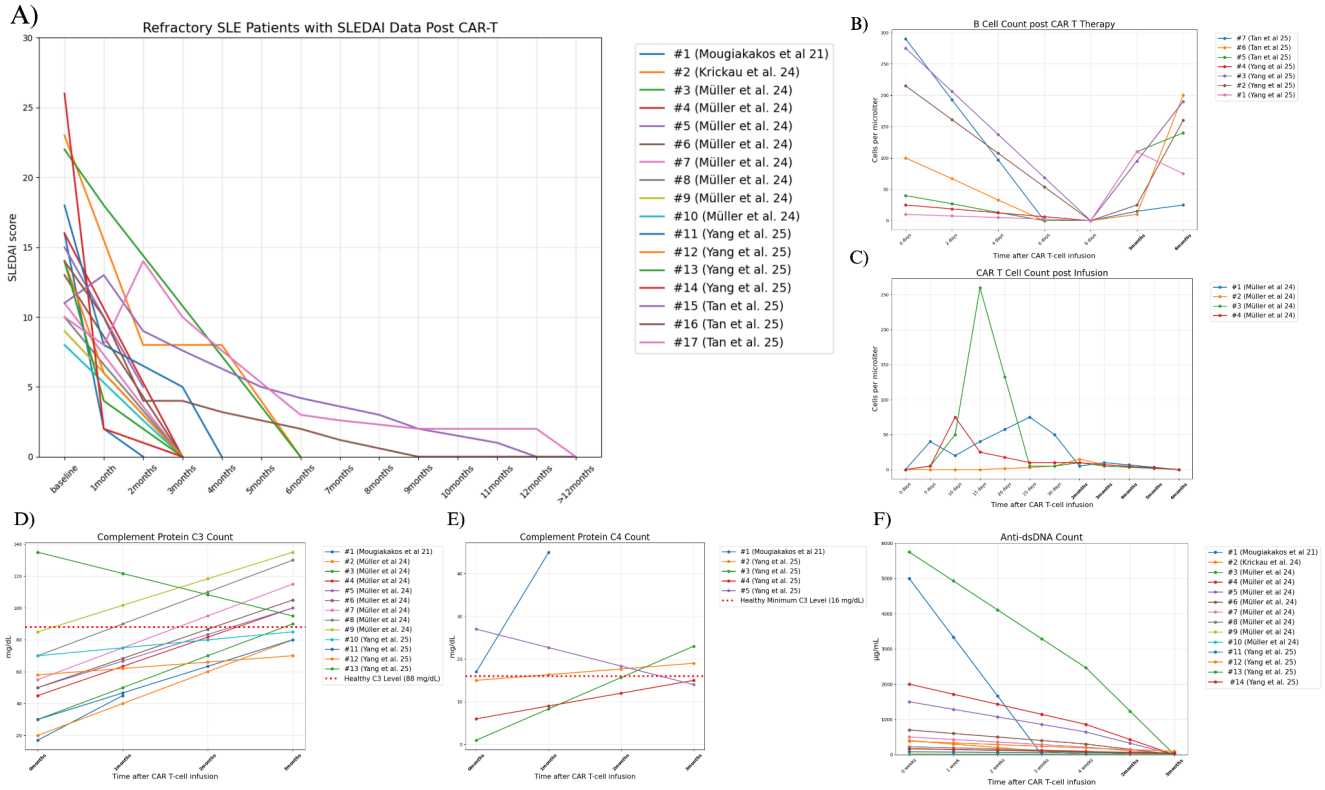

Figure 4. Immunologic and Clinical Data in SLE Patients Treated with CAR T Therapy A) SLEDAI scores from 17 patients are presented from studies with available data B) B cell counts decrease following therapy, and then recover in the following months C) CAR T cell counts initially rise and then decline slowly D-E) C3 and C4 counts increase towards or above healthy minimum levels F) Anti-dsDNA decline markedly in most patients. All values were approximated from published graphs and may not represent exact patient-level data.

Table 1. Summary of Literature Review Studies Results

Author	PATIENTS	Target	CAR-T TYPE	# REMISSION	SIDE EFFECTS
Mougiakakos et al.	1	CD19	Autologous	All	None
Krickau et al.	1	CD19	Autologous	All	Grade 1 CRS
Mackensen et al.	5	CD19	Autologous	All	Grade 1 CRS
Müller et al.	8	CD19	Autologous	All	Grade 1 CRS
Yang et al.	4	CD19	Allogeneic	All	Grade 1 CRS, No GVHD
Tan et al.	4	CD19	Allogeneic	3 (1 patient drop out)	No GVHD
Schett et al. 23	8	CD19	Autologous	All	None
Hagen et al.	18	CD19	Autologous	All	11 Grade 1 CRS, 1 Grade 2 CRS, 3 neutropenia, 2 uncomplicated pneumonia
Schett et al. 24	6	CD19	Autologous	All	Grade 1 & 2 CRS
Nunez et al.	6	CD19	Autologous	All	None
Yuan et al.	12	CD19	Autologous	11	1 Mild UTI

Discussion

Remarkably, we showed that patients with SLE who received a single dose of CAR T therapy almost universally had improvement of laboratory values (C3, C4, proteinuria) and symptom severity (SLEDAI score), which showcases the potential impact of CD19 CAR T cells

as effective in depleting auto-reactive B cells and restoration of naive B cells. This conclusion was reached in the majority of studies included in the review which indicates a promising and extraordinarily innovative approach to SLE treatment. Compared to current treatments which have relatively low rates of remission compared to placebo, CAR T has the ability to induce drug-free remission through a targeted mechanism, thereby reducing the risk of relapse and side effects.

Two interesting studies measured several vaccine-associated antibodies, such as hepatitis B, measles, mumps, rubella, tetanus, pneumococcal, and autoantibodies associated with SLE before and after CAR-T therapy [30, 32]. While both studies showed depletion of most SLE associated autoantibodies, vaccine-associated antibodies levels remained stable [30, 32, 37]. This observation may be due to continuous antigen exposure supporting the formation of CD19+ plasma blasts from naive B cells and CD19+ short-lived plasma cells (SLPC), both of which are short-lived B cells. In the presence of continuous antigen exposure, naive B cells are inhibited from maturing into long-lived plasma cells (LLPC) whereas antigen stimulation for a limited duration allows the naive B cells to mature into LLPCs which lack CD19+ co-receptors on their surface and are unaffected by CAR-T therapy and will then produce the vaccine-associated antibodies. Therefore, CD19 Cart T-cell therapy eliminates the autoreactive CD19+ B cells and its associated autoantibodies while preserving the immunization status of patients.

Side Effects

The majority of the patients, post-CAR T therapy, developed grade 1 CRS with a single patient developing grade 2 CRS but was eventually resolved without complication [26-36]. Three patients developed neutropenia which was reversed quickly with G-CSF treatment. Two

patients developed uncomplicated pneumonia which were successfully treated with antibiotics [33]. One patient developed a mild urinary tract infection [36]. While minor toxicities are common, none of the patients who completed the minimum of 3 months follow up developed any serious adverse effects or complications including ICANS [26-36]. Additionally, none of the patients treated with allogeneic CAR T therapy developed GVHD [30, 31].

Autologous and Allogeneic CAR-T

Allogeneic CAR T therapy offers a potential alternative treatment strategy with unique risks like rejection and GVHD discussed above (Figure 2B). However, in order to prevent these rejection risks, CAR T was genetically modified in the study conducted by Yang et al using CRISPR-Cas9, a gene-editing technology which allows for the precise removal, addition, or alteration of specific genes or parts of the DNA. TRAC, HLA-A, HLA-B, and PDCD1 genes were disrupted before reinfusion. Disrupting the TRAC gene interferes in the formation of the constant region of the alpha chain of the T cell receptor (TCR), preventing the proper formation of the TCR, which would help to prevent GVHD. Disrupting HLA-A and HLA-B genes interferes with the formation of Class I MHC cell surface molecules which would help prevent the recipient's immune system from rejecting the donor CAR T-cells. Lastly, disrupting the PDCD1 gene prevents the Programmed Cell Death Protein 1 (PD-1) formation, which is a cell surface receptor on T and B cells that down-regulates the immune system by suppressing T-cell inflammatory activity [30]. Disrupting this gene increases CAR T cell potency and persistence. The result from Yang et al. shows successful treatment outcomes with allogeneic CAR T. All four patients in the study achieved remission by 3 months and none of the patients experienced any severe adverse effects. Tan et al. which used allogeneic CAR T cells in its study did not

mention which method was used to genetically modify allogeneic CAR T cells. However, the results from this study were not as positive as in Yang et al. Tan et al. found that one patient dropped out of the study due to unresolved thrombocytopenia, and took six months for one patient and greater than 12 months for the other two patients to reach remission. The difference in time to achieve remission between the two studies may have to do with the genetic modification technique used and which genes were modified. Further information is needed to make that determination. However, both of these studies indicate that allogeneic CAR T can potentially be a feasible option that is both safe and effective [31]. If future studies can corroborate similar results for allogeneic CAR-T therapy, then it would have significantly positive implications in helping to bring down costs, the number of required visits, resources, and allow for greater access for a wider patient population.

Limitations

Pooling data from case and pilot studies have inherent weaknesses due to inconsistent designs and lack of statistical power. They are also known to have publication bias, where studies with successful results tend to get published more frequently. Therefore, definitive conclusions cannot be made from the pooled results from this review. However, case and pilot studies are an important initial phase of a novel treatment intervention. They can help determine the feasibility of a new treatment, aid in the design, and identify modifications that may be needed if a larger study is to be conducted [39, 40]

Conclusion

Given that the results of this study are not from randomized controlled clinical studies, the findings cannot be viewed as definitive. However, several trends from the data can be seen from this review. The majority of patients after receiving a single infusion of CAR T therapy improved significantly both serologically and clinically. Most patients reached remission status by 3 months. Furthermore, a significant portion of them were able to achieve a “drug-free” remission. This is not commonly seen with the currently available treatment options where remission has been so elusive. None of the 71 patients that had completed the minimum 3 month follow-up experienced any serious adverse effects or complications. Unlike current treatments which fail to target specific immune cells, CAR T-cell therapy specifically eliminates autoreactive CD19+ B cells, which play an important and central role in the pathophysiology of SLE. This review confirms the understanding of CD19+ cells playing a central role in SLE pathophysiology. The efficacy and safety information from this review shows that CD19 CAR T-cell therapy, including the allogeneic form, are feasible treatment options from a safety and efficacy standpoint for refractory SLE, and it may be able to fill an important unmet need for patients who are suffering from severe disease, experiencing progressive organ damage, and are refractory to the currently available treatment options. Randomized controlled studies are needed to corroborate these results. In addition, a long-term post CAR T treatment follow-up study is needed to assess the long-term remission outcomes and adverse effects.

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